

TIC 318566745

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 18.738148 d (ALIASED, see below)
-> corrected period: 37.476297 d
Epoch (t0) 1332.3398 BTJD
Duration 0.1200 d
Depth 3.112 % (31,125 ppm)
SNR 381.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

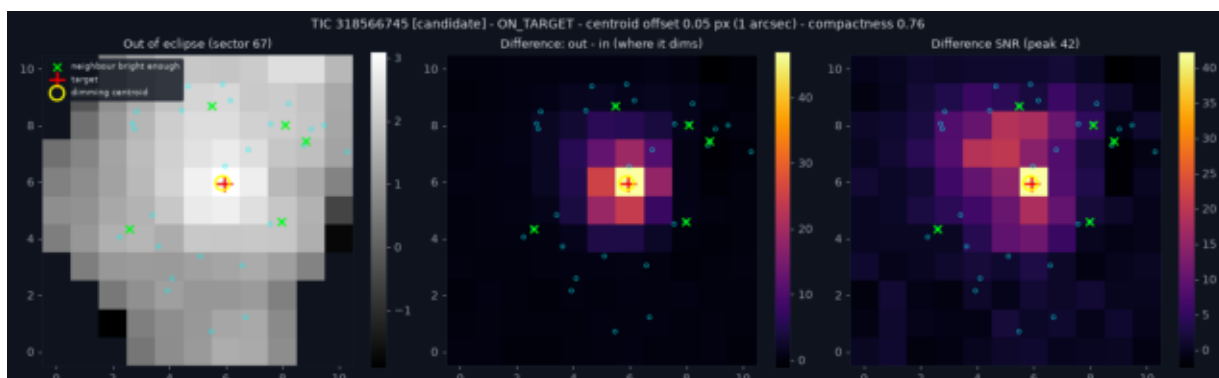
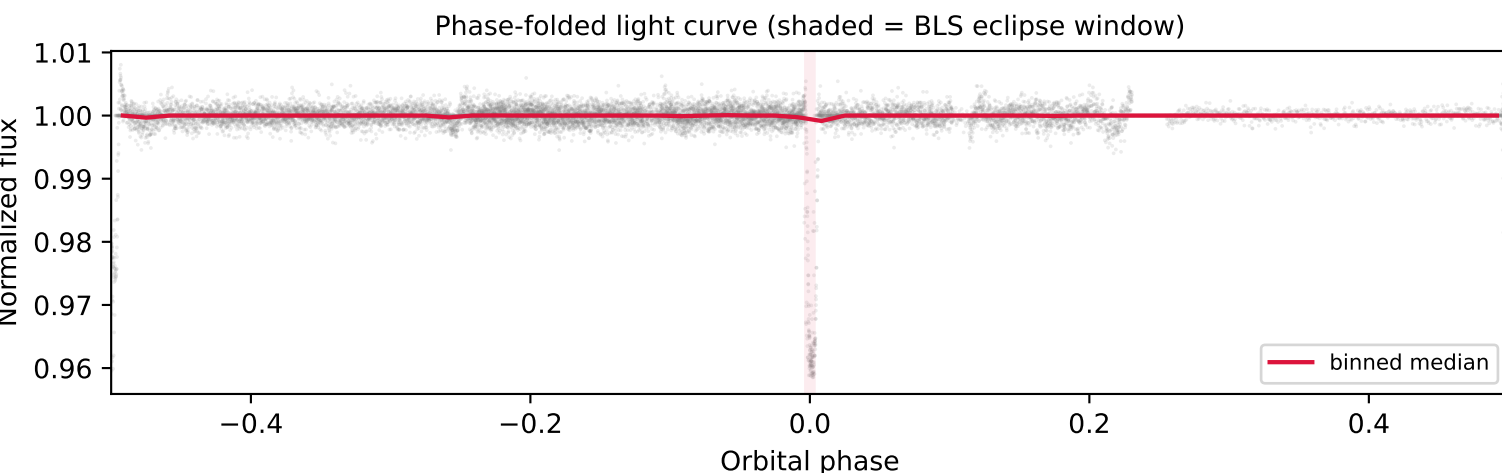
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.17 (gate ≥ 0.50)
Robust depth sigma 364.9 (gate ≥ 7.0)
Odd/even z-score 3.24 (ALIAS FLAGGED)
Secondary eclipse detected, phase -0.47, 5.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 67 light-curve dip in this sector: 171.3 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 42.22 (gate ≥ 5.0) Compactness 0.76 (gate ≥ 0.50)
Dimming centroid offset 0.05 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 318566743
2.50 px away, dTmag +2.84 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.24 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 37.476297 d, in any write-up.